The den limit

the formula is exact; $\gamma < 1$ is a census; den_∞ is not constructed

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Abstract

Round 422 left $\limsup \Sigma < 2 - \mathrm{den}_\infty$ conditional on a constructed $\mathrm{den}_\infty < 2$. The border formula $\mathrm{den} = 1 + \gamma - v_t \cdot s$ is exact ($\gamma = \|b\|^2/B_w$, $B_w = S_{N-2} + 5/7$). $\gamma < 1$ and $v_t \cdot s > 0$ hold on the named census, so den < 2 with gap at least 0.348. $\|b\|^2$ is not a subsum of S. No den_∞ is constructed. No RH claim.

Status. Lemma 1 is **reduced** (hierarchy (iii), with a named (ii)-shaped census). The formula is a finite theorem (Proposition 1). Ausgang `FORMULA_EXACT / GAMMA_LT1_CENSUS / LIMIT_OPEN / SUBSUM_REFUTED`. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (den limit)**.** *On the selected sequence $a_k = 2^k$, $\mathrm{den}_k \to \mathrm{den}_\infty < 2$ with den_∞ source-pure (or at least $\limsup \mathrm{den}_k \le \bar{d} < 2$ with $\bar{d}$ explicit).*

Verdict: REDUZIERT. Hierarchy (i) — a constructed limit — is not reached. Hierarchy (ii) — $\gamma \le \bar{\gamma}$ as a chain theorem — is a universal census $\gamma < 1$, not a SATZ; $\|b\|^2$ is not a subsum of the ρ-series. Hierarchy (iii): on core-42 plus EXT plus selected, den $\le 1.652 < 1.66$ and the gap to 2 is at least 0.348.

2 The formula

Proposition 1 (border normalisation)**.** $\mathrm{den} = 1 + \gamma - v_t \cdot s$ *with* $\gamma = \|b\|^2/B_w$ *and* $B_w = S_{N-2} + \frac{5}{7}$*. Over $\mathbb{Q}$, $\gamma = \frac{2}{3} < 1$, $v_t \cdot s = \frac{1}{6} > 0$ gives* $\mathrm{den} = \frac{3}{2} < 2$*. Even $v_t \cdot s = 0$ would give $1 + \gamma = \frac{5}{3} < 2$. At $k_z = 9$,* $\mathrm{den} = 1.60111$*, $\gamma = 0.6778$, $v_t \cdot s = 0.0767$, $B_w = 8.382$.*

Thus den < 2 follows from the formula as soon as $\gamma < 1$. The correction $v_t \cdot s$ only widens the gap.

3 Anatomy of γ

Selected (live $k = 4, 5, 6, 7, 9$; $k = 8$ pinned):

k	den	γ	$v_t \cdot s$	$\|b\|^2$	B_w
4	1.601	0.678	0.077	5.68	8.38
5	1.651	0.712	0.061	4.58	6.42
6	1.524	0.609	0.085	6.30	10.35
7	1.518	0.604	0.086	7.77	12.87
9	1.482	0.576	0.093	10.88	18.89

Both $\|b\|^2$ and B_w grow. The last-increment ratios $\Delta\|b\|^2/\Delta B_w$ are 0.58 and 0.52 — a Stolz–Cesàro *hint*, not a theorem. $\rho_{N-1} \to 0$ on selected; the ρ_0 share of S falls $0.46 \to 0.26$; S still grows because more terms arrive. γ wanders in $[0.576, 0.712]$ on selected and $[0.548, 0.724]$ on core-42. No γ_∞ is constructed.

Prefix $\|b_{\le t}\|^2/S_t$ falls $0.92 \to 0.73$ ($k = 4$) and $0.91 \to 0.65$ ($k = 6$). So $\|b\|^2$ is *not* a subsum of the ρ-series. Dropping $\frac{5}{7}$ from B_w still leaves $\|b\|^2/S_{N-2} < 1$ (selected max 0.80).

Core-42: $\gamma < 1$ on 42/42, $v_t \cdot s > 0$ on 42/42 ($[0.067, 0.096]$). EXT 6/6 the same. If $\gamma < 1$ were a theorem, den < 2 would be a theorem via Proposition 1. It is not proved in this round.

4 The bound that is actually in hand

On the named census, $\gamma \le 0.724$ and $v_t \cdot s \ge 0.067$, so

$$\text{den} \;\le\; 1 + 0.724 - 0.067 \;=\; 1.657 \;<\; 2.$$

The measured maximum is 1.652; the gap to 2 is at least 0.348. An honest explicit $\bar{d}$ is the census number 1.66, not a source-pure constant.

5 Kills

Dead vacuous χ: den $= 1.546$ and 1.597, both < 2. Death is Σ/razor occupation (r422), not den. Drop $v_t \cdot s$: $1 + \gamma$ stays < 2 on the census. Drop $\frac{5}{7}$: γ still < 1. Scramble $\text{nneg}(A_0) = 21$. Drop-border: no den scalar.

6 What is a theorem, what is a census

Proposition 1 is a theorem. Lemma 1 is not: den_∞ is not built, $\gamma < 1$ is not a chain identity, and $\|b\|^2 \le B_w$ is observed rather than proved. The cofinal edge still needs a constructed den_∞ or a proof of $\gamma < 1$. No RH claim.